



Birla Vishvakarma Mahavidyalaya Engineering College

The Robotics Society, India Student chapter

Report on

Expert Talk on "Latest Technology in
Robotics and Related Innovations"

Date and Day: 19-02-2025, Wednesday

BIRLA VISHVAKARMA MAHAVIDYALAYA
[AN AUTONOMOUS CVM INSTITUTION WITH NAAC 'A']
VALLABH VIDYANAGAR, ANAND, GUJARAT
BVM TRS STUDENT CHAPTER ORGANIZES

EXPERT TALK ON
"LATEST TECHNOLOGY IN ROBOTICS AND RELATED INNOVATIONS"

KEYNOTE SPEAKER
ER. KRISHAN GOTEWAL
DIVISION HEAD, RHRTD, IPR

KEYNOTE SPEAKER
ER. NAVEEN RASTOGI
SCIENTIFIC OFFICER, IPR



START AT
03 PM - 05 PM



WEDNESDAY ,
19 FEB 2025



PARTICIPATION
CERTIFICATE

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limited seats only

1. Objective:

The primary objective of the talk is to introduce students to the latest trends in robotics and automation, explore real-world applications of emerging robotic technologies, encourage interaction with industry experts and student, and inspire innovation and research in robotics-related fields.

2. Details of the Expert Talk:

Date and Day: 19-02-2025, Wednesday.

Participants:

- 35 Students member from the branches Electronics Engineering, Mechanical Engineering, Electrical Engineering, Electronics & Communication Engineering
- 4 Faculty members from Electronics Engineering, Mechanical Engineering, Electronics & Communication Engineering

3. Expert Talk Details

Experts from Institute for Plasma Research, Gandhinagar Mr. Krishan Kumar Gotewal, Scientific Officer – H and division head of the Remote Handling and Robotics Technology Development Division (RHRTD) at the Institute for Plasma Research and Mr. Naveen Rastogi is working as Scientific Officer – F at the Institute for Plasma Research, Gandhinagar delivered a talk. The session covered insights into advancements in robotics, particularly in Remote Handling (RH) systems for fusion devices. The session covered various aspects of robotics, including types of robots, applications in fusion research, and the role of virtual reality (VR) in designing RH-compatible plant systems.

Mr. Naveen Rastogi Discussed about

Types of Robots:

The talk began with an overview of different types of robots, categorized based on their applications and structural configurations. These included:

- **Industrial Robots:** Used in manufacturing, assembly, and quality control.
- **Autonomous Robots:** Capable of performing tasks without human intervention.
- **Humanoid Robots:** Designed to mimic human actions and behaviors.

- **Medical Robots:** Utilized in surgeries, rehabilitation, and patient care.
- **Service Robots:** Employed in household and commercial services.
- **Space and Defense Robots:** Used in planetary exploration and military applications.

Mr. Krishan Kumar Gotewal, Discussed about

What is RH (Remote Handling)? Remote Handling (RH) refers to the use of robotic systems to manipulate objects in hazardous or inaccessible environments. It is a crucial technology in nuclear fusion research, where human intervention is limited due to high radiation levels and extreme operational conditions.

Why is RH Needed in Fusion Devices? RH is essential in fusion devices for the following reasons:

- Ensuring the safety of personnel by eliminating direct exposure to radiation.
- Enabling maintenance and replacement of critical components without shutting down the reactor.
- Enhancing operational efficiency and reliability in high-temperature and high-energy environments.
- Supporting the longevity and sustainability of fusion power plants.

Typical RH Systems in Various TOKAMAKs: The session discussed RH systems implemented in major TOKAMAKs, including:

- **JET (Joint European Torus):** Features robotic arms for remote maintenance.
- **ITER (International Thermonuclear Experimental Reactor):** Utilizes sophisticated RH systems for diagnostics and component replacement.
- **DIII-D (San Diego):** Incorporates RH tools for plasma diagnostics and tokamak maintenance.
- **JT-60SA (Japan):** Employs robotic solutions for high-precision interventions.

Design for Maintenance (DFM) During System Design Phase: A key aspect of RH is the Design for Maintenance (DFM) approach, which ensures that:

- Components are designed for easy accessibility and replacement.
- Robotic interfaces are integrated from the early design phase.
- Modular designs are implemented for faster repairs and upgrades.

- Maintenance strategies are validated through simulations before deployment.

Role of VR in Designing RH-Compatible Plant Systems: Virtual Reality (VR) plays a crucial role in the development of RH-compatible plant systems by:

- Providing an immersive environment for design validation and optimization.
- Enabling remote training and skill development for RH operators.
- Simulating real-time maintenance scenarios to test RH feasibility.
- Enhancing human-robot collaboration through interactive planning.

RHRTD Activities Highlights: The talk concluded with highlights from recent Remote Handling Research and Technology Development (RHRTD) activities, including:

- Advances in robotic dexterity and AI-driven decision-making.
- Development of lightweight and radiation-resistant robotic components.
- Innovations in teleoperation and haptic feedback systems.
- Successful case studies of RH applications in fusion research.

4. Outcomes:

- Participants gained insights into the latest technological advancements in robotics.
- Enhanced awareness about research and career opportunities in the field.
- Interactive Q&A session helped address specific doubts and interests of attendees.
- Encouraged students to explore robotics-related projects and competitions.

5. Conclusion: The expert talk provided a comprehensive overview of the latest developments in robotics and RH systems, emphasizing their critical role in fusion research and industrial applications. The insights shared by the expert will aid in fostering further research and development in this domain, paving the way for more efficient and safe operations in high-risk environments.

6. Glimpses:

